Introduction to Causal Inference for Software Engineering

A 90-minute hands-on tutorial

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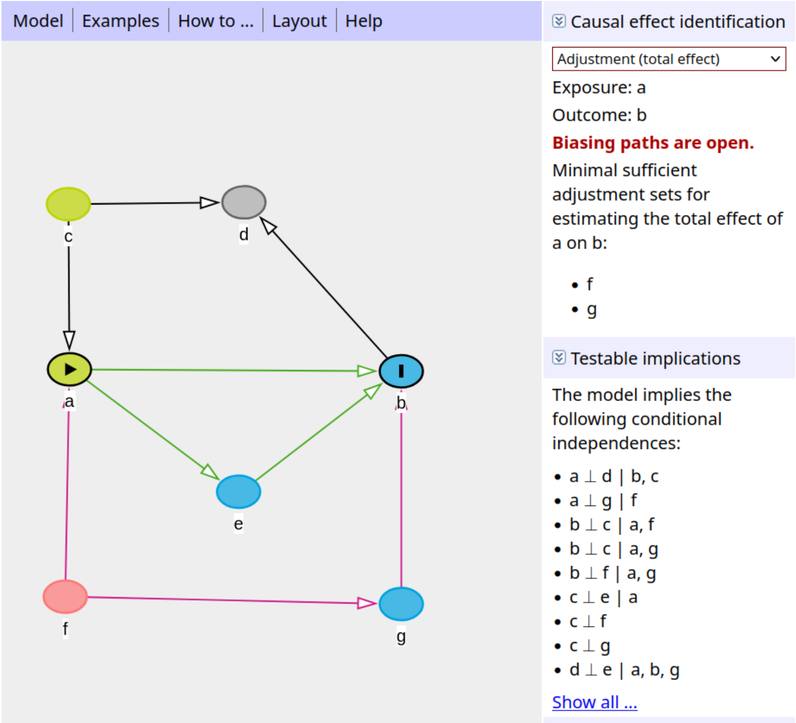
Section 4

Exercises: Identification, adjustment sets and conditional independence

A guides tour through DAGitty

- DAGitty is a browser-based tool to create and analyse DAGs
- It helps causal inference by automatically identifiying necessary adjustment sets.
- Based on the graphical causal model, it provides ***conditional independence criteria*** that can be used to test models.
- Access it here: https://dagitty.net/



Textor et al., Robust causal inference using directed acyclic graphs: the R package 'dagitty'. *International Journal of Epidemiology* 45(6):1887-1894, 2016.

Live Demo

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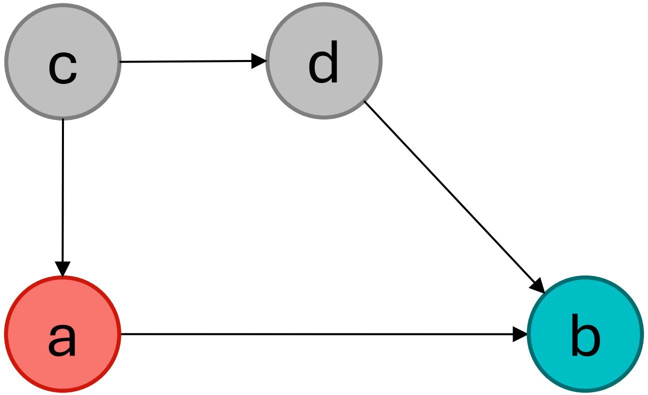
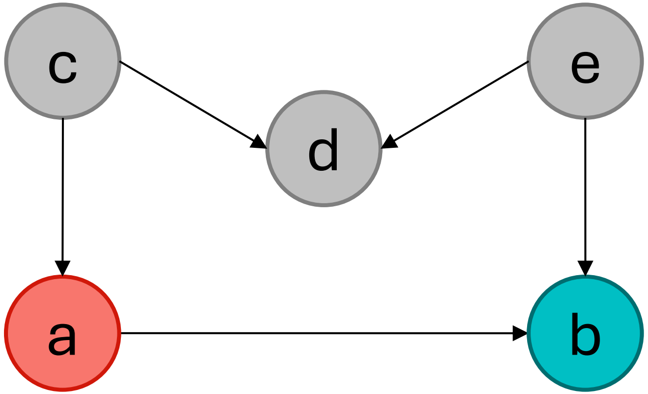
Problem 1

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



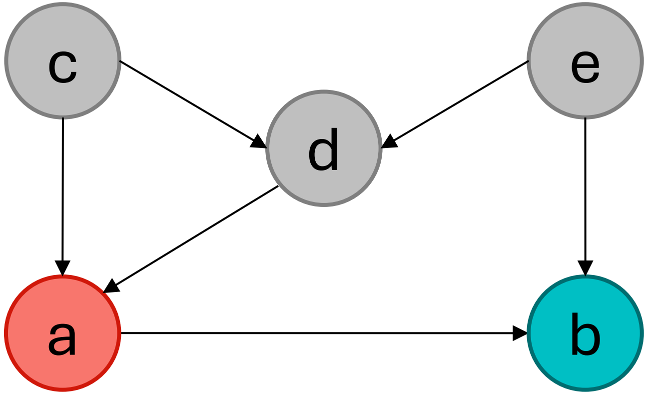
Problem 2

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



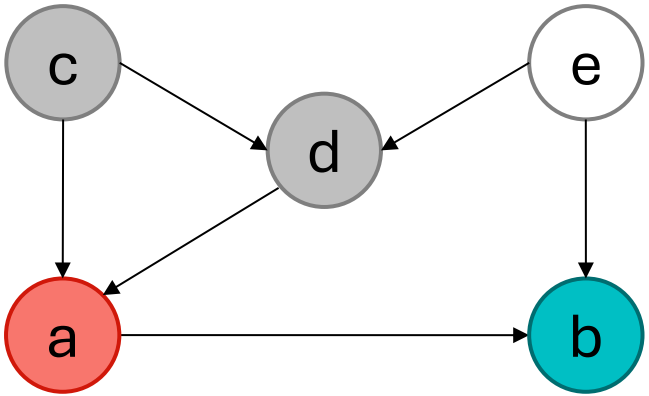
Problem 3

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



Problem 3 more interesting

- Determine the **adjustment set** of variables to block any backdoor causal path for the estimation of the direct effect of a on b.



Problem 4: Does an automated testing framework (X) reduce post-relase defects (Y)?

A software quality team studies the effect of the introduction of an automated testing framework on post-relase defects of a large system. They know:

1. Experienced developers are more likely to adopt the framework and to write less defect-prone code.
2. The framework lowers defects directly, and also by raising test coverage.
3. More complex modules tend to have more post-release defects.
4. Both the framework and module complexity raise the number of issues logged.

Draw a DAG including the variables: Autom. Testing, Post-Release Defects, Developer Experience, Test Coverage, Logged Issues, Code Complexity.

Identify:

- Which variables are confounders;
- Which are mediators;
- What is the minimal adjustment set to estimate **the direct and total effect** of automated testing on post-release defects.